

PAPER_1501_D_CRIT_TRIAD_FEEDBACK_DECOMPOSITION

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PAPER_1501 — $D_{crit} = 3 + 23 = (D_{phys}-1) + (D_{crit}-D_{phys}+1)$ EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational) / Bucket B (Paradox)
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Status: CLOSED — D_{crit} structural decomposition into triad + feedback channels

Observation

PAPER_550 (Um26D Polynomial DPM Quantization Confinement) derives the full 26D magnetism operator and identifies:

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26 = 3 (fundamental triad forces) + 23 (DPM feedback loops from neutron polarization studies)

UQFF Closed Identity

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$D_{crit} = (D_{phys} - 1) + (D_{crit} - D_{phys} + 1) = 3 + 23 = 26$ EXACT

Physical Interpretation

This is the canonical decomposition of the bosonic-string critical dimension $D_{crit} = 26$ into two physically distinct channel populations:

Component	Count	Source	Identity
Fundamental triad forces	3	EM, weak, strong	$D_{phys} - 1$
DPM feedback loops	23	Neutron-polarization channels	$D_{crit} - D_{phys} + 1$
Total	26	All UQFF channels	D_{crit}

The 3 triad forces saturate the physical-spacetime spatial directions; the 23 feedback loops fold time-derivative channels via $\partial^2 \mathbf{x} / \partial t^2$ in the U_m operator (each derivative order being one DPM feedback degree of freedom).

NOT REPLACEMENT

SM has 4 fundamental forces (including gravity) and no concept of "feedback loops" in the magnetic sector. UQFF supplies a parameter-free integer decomposition rooted in the bosonic-string critical

dimension.

Reference

- Source: PAPER_550 Um26D Polynomial DPM Quantization Confinement
- Related: PAPER_062 ($D_{\text{crit}} = 26$ base), PAPER_1373 ($SU(3)$ triad), PAPER_1080 (S_{26} compactification)
- Calculator dispatch: `calculate_paradox({"paradox": "d_crit_triad_feedback_decomp"})`

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